

Weekly Diary – Week 10

Internship Organization: Vikaspedia

Duration: 22/03/2026 – 28/03/2026

Objective of the Week

The main objective for this week was to enhance the overall performance of the system, implement effective error handling mechanisms, and complete the final stages of documentation and testing. The focus was on making the system more efficient, stable, and user-friendly while ensuring that it meets all functional and technical requirements before submission.

Date – 22 March

On this day, performance optimization was carried out by modifying the communication strategy. Instead of transmitting the entire document data during each update, incremental changes were sent between users. This approach significantly reduced bandwidth consumption and improved the efficiency of real-time synchronization. As a result, the system became faster and more scalable.

Date – 23 March

Efforts were made to further reduce latency in real-time communication. The data exchange process was carefully analyzed and fine-tuned to ensure quicker response times. Adjustments were implemented in both client-side and server-side communication flow to minimize delays, thereby improving the overall user experience during collaborative editing.

Date – 24 March

Error handling mechanisms were implemented to manage unexpected WebSocket disconnections. The system was enhanced with automatic reconnection logic to ensure that users could reconnect seamlessly without losing their session progress. Additionally, appropriate notifications were displayed to inform users about connection status, improving usability and reliability.

Date – 25 March

Comprehensive testing was conducted under various conditions, including multiple concurrent users and different network scenarios. These tests were essential to evaluate system stability and ensure that

communication remained consistent even under fluctuating network conditions. The results confirmed reliable performance across different environments.

Date – 26 March

The codebase was refactored to improve readability, structure, and maintainability. Redundant code segments were removed, and proper coding standards were followed. Meaningful comments were added throughout the code to enhance clarity, making it easier for future developers to understand and modify the system.

Date – 27 March

Complete project documentation was prepared on this day. This included detailed installation guidelines, system architecture explanations, and clear instructions on how to use the application. The documentation was structured in a way that ensures ease of understanding for both technical and non-technical users.

Date – 28 March

Final testing and debugging of the entire system were performed. All modules were thoroughly verified to ensure that they function correctly and meet the defined requirements. Any remaining issues were identified and resolved, ensuring that the system was stable, efficient, and ready for submission.